Innovation Brief

1. Elevator Pitch: Pitch your innovation, sharing its essence, impact, customers and business potential. (50 words)-Sahasra

AeroFold is an origami-inspired, nitinol deorbit module that integrates seamlessly into satellites, enabling them to re-enter Earth's atmosphere in years, not decades. It's inexpensive, jointless, and robust, making it reliable for operators to meet deorbit regulations and reduce space debris.

2. Team: How did your team form? What role will each team member play? What motivated you to make this innovation? What special capabilities, resources or experiences do your team members bring? (150 words)- abhi

What can we design to solve problems today that also prevents tomorrow's? The desire to explore this question is what brought us together. Sahasra Beerala's experience/role is centered around business and project management. Reshwant Borra's experience/role is tailored to coding and website design. Abhirami Soundararajan's experience/role is geared towards engineering, market research, and resource allocation. Armani Njogu's experience/role is mainly in engineering to develop the CAD model and prototype. Siddharth Mohan's experience/role encompasses material selection and satellite-Aerofold integration. Collectively, each team member brings experience in CAD, structural, and material design, business strategy, and market research. After realizing the growing issue of space debris and the lack of affordable and reliable deorbit solutions for small satellites, we found our innovation's motivation, which is to make space safer. This led us to create a design that maintains environmental responsibility and regulatory pressure in Low Earth Orbit, while being passive and low-mass.

3. Opportunity: What issue or pain point does your innovation address? (300 words)- abhi

There are approximately 40,000 space debris objects that have been located and tracked. Additionally, this number continues to rise as more space debris is being discovered. This underlying issue can create catastrophic risks such as satellite collisions, and cascading debris events, which are more commonly known as the Kessler Syndrome. Our innovation addresses the urgent and growing problem of space debris accumulation specifically located in Low Earth Orbit (LEO). If space debris is unmanaged, it could potentially lead to a depreciation of the long-term sustainability of the LEO.

Current deorbit technology that is available, such as drag sails, has numerous hinges and deployment booms leading to it becoming complex. Moreover, the current deorbit technology is either fragile, costly, or difficult to integrate into satellites and spacecraft that contain small spaces. Many satellites, such as CubeSats, lack the correct disposal system, causing them to contribute to space debris. These limitations show what problems satellite operators must overcome, which include balancing cost along with mission performance.

Our origami-inspired deorbit systems address these gaps by using a tessellated deployable structure that is able to fit into tight spaces and unfold into a large rigid drag sail that causes increased aerodynamic drag without encompassing complexity. Its compactness, robustness, low mass, affordability, and reliability make it well-suited for small satellites/spacecrafts. Satellite manufacturers can integrate our innovation as a standard part, while following regulations and contributing to preserving the orbital environment for future missions.

4. Innovation: Describe your innovation, its design and your technology. How does it work? What is new or proprietary about the innovation? How does it meet needs and resolve pain points? What impact does your innovation create for individual users and for humankind? Describe this qualitatively and quantitatively. How can new or proprietary aspects be protected and made valuable by one or more methods such as a patent, trade secret, copyright or otherwise competitively defensible configuration? (750 words) -sid

5. Validation and Progress: How have you validated your innovation, technology or processes? What progress have you made in developing your innovation? [make sure to include future steps if possible] (450 words)- rishi
We have been able to prove the concept of AeroFold through a series of intense designs and logical analyses and simulations. At this stage of our development, it is necessary for us to be convinced that as a concept, AeroFold is technologically viable and has potential for effectively combating debris in orbit before we proceed to physical demonstrations.

Our validation procedures thus originate from the design as well as geometry of our deployable structure. We have designed various origami tessellations from two-dimensional concepts to three-dimensional representations utilizing the tool of Autodesk Inventor. Through various iterations, we tested the fold, store, and deployment characteristics of various tessellations, especially under the compact constraints of a satellite. After a series of iterations, we concluded that the most optimal tessellation design is the square tessellation. Square tessellations have a natural aptitude to surface as a box, which happens to have a certain compatibility with satellites, especially CubeSats.

After forming the tessellation pattern, we then turned to testing the aerodynamic functionality by employing OpenFOAM to model drag characteristics in low orbit around the earth as well as testing the effect of AeroFold once launched. In order to have as realistic results as possible for testing purposes, AeroFold was designed using a model of the SpaceX Starlink V2 satellite, which is one of the most popular types of satellites currently in orbit around the planet. In both cases, we were able to conclude that AeroFold would significantly increase the cross-sectional area of satellites to such an extent that they can deorbit within five years after deployment. This is consistent with current rules for satellites to deorbit within twenty-five years after fulfilling their mission requirements.

Alongside the simulations, there have been a lot of advancements in the engineering process. We have now completed the system design, decided on the deployment plan, and chosen the materials for the system. These are aluminized PET (Mylar) and Dyneema for the deployable surface, as well as coilable CFRP tape spring booms for the structure. These materials have been selected in order to create a sturdier system that has a better lifespan than drag sails made of thin film. Moreover, a patent application has also been submitted for the deployment system of AeroFold, which has an origami design.

The next course of action revolves around physical development. This will include the manufacturing of the first deployable prototype and some laboratory testing. The outcomes will also be compared against our simulation results. This will allow us to refine our design even further and allow us to be better assured about the capabilities of AeroFold in a "real-world" environment.

6. Market: Describe your customers and your target segments. What is important to them? What is the size of the opportunity? Is the buyer or payer different from the customer in this market? Describe the industry ecosystem. (300 words)-Sahasra

The industry ecosystem of our product is composed of satellite operators, bus manufacturers, deployer/launch providers, subsystem suppliers, and regulators/insurers. AeroFold fits as a standardized bolt-on module, selected during bus design and integrated during assembly.

As a result, in our primary market, smallsat deorbit hardware, the buyers and payers are different from our customers. First, our customers specify requirements for satellite and deorbit modules, selecting AeroFold as their option. Then, the satellite bus manufacturers will purchase AeroFold in bulk to integrate into bus designs. Deployer/launch providers act as our secondary buyers, requiring compliance for launch manifests.

Our primary customer, commercial LEO small satellite operators, have pressure to standardize identical deorbit solutions across entire constellations, achieve fast sales, and meet licensing requirements. As a result, regulatory compliance, reliability, cost, minimal mass/volume, and predictable end-of-life are factors that they prioritize, especially since the international 25-year deorbit rule and emerging 5-year rule. Due to this immediate need and 50-500+ satellite launches each year, operators place bulk orders on deorbit modules, causing repeated business and making them the ideal biggest customers for our product.

Our secondary customers are university and research teams. Although they have smaller budgets, the market is made up of high-volume individual projects and prototype opportunities, providing early credibility to our product.

Combining customers and buyers, the size of this market is significant. Over 5,000 small satellites launch annually, with 70-80% in LEO where deorbit hardware is required. Currently, 10–20% (~500–1,000 units/year) require dedicated deorbit modules, representing a $250M–$2B annual addressable market at $500–$2,000 per unit. Emerging 5-year LEO deorbit rules will drive adoption to over 50% of launches, with commercial constellations representing the highest-volume buyers.

To this growing market, AeroFold provides a scalable, reliable, cost-effective, and sustainable option compared to our competitors.

7. Competition: What competes with your innovation, and how does your innovation compare? What are the advantages and disadvantages of your innovation? What is your positioning? (300 words)-Sahasra

Our primary competitors are thin-film drag sails, specifically Spinnaker Series and ADEO Series, which are two leading solutions in the market. Although they utilize similar physics as AeroFold, these drag sails use fragile/thin polymer membranes with metal coating, making them prone to tearing, wrinkling, or incomplete deployment. Furthermore, they rely heavily on hinges, booms, or complex deployment mechanisms that require a large volume, causing jams and limited control over sail geometry.

Onboarding propulsion systems, including chemical/electric thrusters, have high effectiveness due to controlled deorbit burns and trajectory management. However, due to power requirements and materials, the systems have high prices, mass, and operational complexity. For small satellites, propulsion is over-engineered for end-of-life disposal.

Despite these options, some operators still chose not to insert any deorbit system, relying on the natural drag, accepting a long deorbit. Although this avoids complexity, it increases collision risks, causing increased costs for avoidance maneuvers, breaking regulations, and licensing.

Compared to these competitors, AeroFold has advantages due to its origami tessellation designed with shape-memory alloy nitinol. Unlike conventional drag sails that use fragile membranes, AeroFold's material is trained to unfold into a precise geometry when heated. Even if damaged during launch or in space by other debris, the material will return back to its original shape. Furthermore, the origami-inspired square tessellation shape allows for a high expansion ratio from a compact volume . Overall, our solution is inexpensive, with reduced mass/volume, and no operational complexity. While thermal conditions might seem like a problem, AeroFold uses insulation and temperature regulations.

AeroFold is positioned as the next-generation passive deorbit module that bridges the gap between fragile thin-film drag sails and heavy/expensive propulsion systems.

8. Go-to-Market: How will you attract and sell to customers? Who are the best initial or pilot customers? Is the market best served through direct sales, distribution, licensing, strategic partnerships or other strategies? (150 words)- armani

Our market strategy starts with focusing on direct engagement with satellite manufacturers as our initial customers. Our pilot customers are small satellite and CubeSat manufacturers that operate in Low Earth Orbit, since they must comply with the 5-year lower earth deorbit rule and with the record breaking more than 390 CubeSats being launched in 2025 there is a large customer base. These customers face increasing pressure to address sustainability which makes them ideal early adopters.

AeroFold will be sold as a component that can be integrated during satellite manufacturing. Sales will be made through direct outreach to satellite manufacturers along with strategic partnerships. We would also market at aerospace industry conferences and expos to increase our products exposure. In addition, we will collaborate with regulatory agencies to establish our solution as a compliant deorbit solution. This strategy allows for customer acquisition, early validation, and scalability across the satellite market.

9. Business Model: What are your key revenues and costs? What are the pricing and costs to deliver one product or service unit? (300 words)- armani

AeroFold would generate revenue through direct product sales, licensing agreements and strategic partnerships. Our main revenue stream would come from selling AeroFold as a component to satellite manufacturers and would be priced per satellite and with over 10,000 of just Starlink satellites currently in orbit and more to follow provides us with a large customer base. Additionally, we will gain revenue from licensing agreements of AeroFold to aerospace companies along with government grants and public funding from organizations with interest in space sustainability. Another revenue opportunity would be customer support.

Our key costs would be materials, manufacturing, testing and integration. The most significant variable cost includes aerospace-grade aluminized PET (Mylar) along with other high-end aerospace-grade materials. Additional cost would be the precision manufacturing required for some of our parts along with assurance testing, packaging, and logistics required to deliver our product. Fixed costs include research and development, engineering labor and prototyping.

The cost to deliver one AeroFold unit is largely from material selection and manufacturing scale. As production volume increases the bulk sourcing of materials and optimization of manufacturing would significantly reduce the per-unit costs. Our pricing strategy aims to balance affordability while providing a high-quality product for our customers while also supporting continued innovation. This business model is financially sustainable, scalable across the numerous satellite platforms and focused on the growing demand for space sustainability.

10. Fundraising: What funds do you need to get started, and how will you use these funds? How much will it cost to develop the product and roll it out? What different sources will you pursue for funding, and why are these a fit?(150 words)- rishi

In order to proceed with planning and development of AeroFold from a simulated concept to a physical implementation, initial funding for prototyping and initial physical testing would be necessary. This initial funding will be used for initial purchases related to aerospace materials such as material for a deployable surface for the structure, which could be aluminized PET (Mylar) and Dyneema, as well as coilable CFRP tape spring booms.

We intend to look for funding through a combination of innovation competitions, research grants, government investment, as well as partnerships with satellite and space-related organizations. Such sources would indeed complement the innovation because AeroFold is decidedly relevant to the increasing issue, whether regulatory or environmental, of space debris mitigation. Investing or receiving research grants is appropriate since the innovation is in line with the public need for responsible space activity, whereas partnerships with satellite manufacturers bring the innovation closer to practical application.

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